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# Manas Sashank Juvvi

## Profile

Controls and Robotics engineer with an MSc from TU Delft and 1.5+ years of full-time R&D experience. Specialised in *autonomous navigation, motion planning, localisation and control* for mobile robots and UAVs, with hands-on experience across *multi-robot systems, scaled autonomous race cars and quadcopter platforms*. Experienced in developing and deploying robotics software in *C++, Python, ROS 2, MATLAB/Simulink and Isaac Lab*, from simulation through hardware deployment.

## Education

- 2024 – 2026 *MSc Systems and Control* *Delft University of Technology*  
*Thesis* (Nov 2025 – Aug 2026): *A Statistical Framework for Counterexample Driven Refinement of Reinforcement Learning Controllers*, supervised by Dr. Manuel Mazo Espinosa.
- Developed a *train-certify-refine* framework for improving *reinforcement learning controllers* for mobile robots under task and safety constraints, combining controller training, refining and *statistical certification* from finite rollout data. Validated on ground-robot and quadcopter.
  - Achieved *44% higher mean task robustness* than the leading baseline (TGPO\*) with *4× lower compute time*.
- Coursework*: Model Predictive Control, Optimization, Machine Learning for Systems and Control, Autonomous Flight of Micro Air Vehicles, Control in Human-Robot Interaction, Modeling of Dynamical Systems
- Jun 2025 *ETH Zürich Robotics Summer School* *ETH Zürich, Geneva*  
Selected for a competitive robotics summer school, where I worked as part of a team to develop and deploy the full autonomy stack of a rough-terrain mobile robot, integrating *perception, localisation, mapping, planning and navigation* to execute a search-and-rescue mission.
- 2019 – 2023 *BTech Mechatronics*, minor in Robotics and Automation *Manipal Institute of Technology*  
CGPA 8.48/10 ■ *Coursework*: Robot Dynamics and Control, Linear Control Theory, Mobile Robots and Path Planning, Machine Vision and Image Processing, Artificial Intelligence

## Experience

- Jan 2023 – Jul 2024 *Research Associate*, Indian Institute of Science *Bangalore, India*  
Formal Control and Autonomous Systems (FOCAS) Lab, PI Dr. Pushpak Jagtap
- Carried out two research projects from problem formulation through *algorithm development and hardware validation*, resulting in first-author papers at ICRA 2024 and IROS 2025 and a co-authored ICCPS 2024 poster.
  - Developed and maintained *C++ and Python robotics software* using ROS 2 for three platforms, a TurtleBot3 fleet, an F1TENTH race car, and a research-grade outdoor rover.
  - Designed and built a *research-grade outdoor mobile robot* from concept to field-tested prototype, covering *electronics, mechanical assembly, low-level control and autonomous navigation*.
  - Mentored 4 junior students on ROS, reinforcement learning and formal control methods.

## Selected Projects

- Jan 2023 – Jun 2024 *Research-Grade Outdoor Mobile Robot* | OAK-D, Jetson Xavier NX, C++, Python, ROS, Arduino
- Developed the *perception pipeline* using an *OAK-D depth camera* for terrain classification and object-triggered behaviour, running onboard a *Jetson Xavier NX*.
  - Implemented *object-triggered behaviour*, detecting specific objects to drive towards them and stop at a defined distance, closing the loop between perception and motion.
  - Developed *low-level control algorithms* using prescribed-performance control, and integrated ROS packages for *autonomous navigation and exploration*.

- Sep 2021 – *e-Yantra Robotics Competition: Strawberry Stacker* ↗ | *Python, OpenCV, ROS, PX4* video ↗
- Sep 2022
- Built a *vision-guided pick-and-place* system for multiple quadcopters with *ArUco marker detection* to locate boxes and *colour/contour-based detection* to locate trucks for loading.
  - Fed detected positions into the *control and navigation pipeline* (ROS, PX4) to guide the drones to pick and place targets. Team reached the *Grand Finals (top 8 of 250+ teams)*.
- Sep 2023 – *Motion Primitive Reinforcement Learning* | *Python, PyTorch, ROS*
- Mar 2024
- Combined *reinforcement learning and sampling-based planning* to adapt robot behaviour to new task specifications *without retraining*.
  - Developed a reusable library of RL motion primitives that transfers across tasks and environments, converting a new specification from a training problem into a planning query.
  - Developed a *Gaussian-process reachability model* linking planning and control. Validated the approach on *TurtleBot3* and *Unitree Go1* in simulation and on hardware.
- Apr 2023 – *Safe Multi-Robot Exploration* | *C++, Python, ROS, multi-agent systems*
- Sep 2023
- Developed a multi-robot exploration stack with provable collision avoidance, using symbolic control for each robot's local controller and control barrier functions to compose them safely.
  - Designed and deployed a *frontier target selection algorithm* that ranks exploration targets using safety margins around unmapped obstacles on a *TurtleBot3 fleet*.
- Jun 2023 – *F1TENTH Autonomous Race Car* | *Python, ROS, MATLAB*
- Jun 2024
- Developed a *data-driven Gaussian-process model* to replace a hand-tuned kinematic model, improving vehicle-state prediction accuracy at high speeds.
  - Developed a *dynamic-obstacle prediction* module that evaluates collision risk across the planner's state space, enabling more aggressive trajectory generation in low-risk regions.

## Publications

- IROS 2025 *Signal Temporal Logic Compliant Co-design of Planning and Control* ↗  
video ↗ **M. S. Juvvi**, T. D. Kurne, V. J. S. Kolathaya, P. Jagtap  
*IEEE/RSJ International Conference on Intelligent Robots and Systems*, Hangzhou, China
- ICRA 2024 *Safe Multi-Robot Exploration using Symbolic Control* ↗  
video ↗ **M. S. Juvvi**, D. S. Sundarsingh, R. Das, P. Jagtap  
*IEEE International Conference on Robotics and Automation*, Yokohama, Japan

## Skills

|                    |                                                                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages          | Python, C++, MATLAB & Simulink, Embedded C, Bash                                                                                                                                                                     |
| Tools & Frameworks | ROS, ROS 2, Gazebo, NVIDIA Isaac Sim & Isaac Lab, PX4, OpenCV, PyTorch, TensorFlow, Docker, Git, CMake, Linux                                                                                                        |
| Domain Expertise   | Autonomous navigation, localisation, motion & trajectory planning, SLAM, vision-based navigation, controls, model predictive control, state estimation, sensor fusion, system identification, reinforcement learning |
| Hardware Platforms | TurtleBot 2/3/4, Crazyflie 2.1, F1TENTH, Unitree Go1, UR5e, Jetson Xavier NX, Raspberry Pi, ESP32                                                                                                                    |

## Extracurricular

- Jun 2021 – *Manipal Institute of Technology* *Manipal, India*
- Oct 2022 *Vice Chairperson*, IEEE Robotics and Automation Society Student Chapter
- Grew the chapter to 50+ active members and organised an industrial robotics lecture series.
  - Compiled educational resources covering programming, robot kinematics, ROS, and machine learning for new club members.
- Jul 2021 – *Manipal Institute of Technology* *Manipal, India*
- Sep 2022 *General Secretary*, IE Mechatronics Students' Chapter
- Founded the Robotics subsystem and mentored new members in programming, linear algebra, robot dynamics and ROS.
  - Organised "Robotron," a hands-on workshop on the UR5e and TurtleBot 2 platforms.